Angles & Parallel Lines Error Analysis TEACHER KEY

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

1. Vertical angles

ORIGINAL PROBLEM

Two lines intersect. One angle is $(3x + 10)°$ and its vertical angle is $(5x - 20)°$. Find x.

INCORRECT WORKED SOLUTION

$(3x + 10) + (5x - 20) = 180$

$8x - 10 = 180$

$8x = 190$

$x = \frac{95}{4}$

A. EXPLAIN THE MISTAKE (1 POINT)

Vertical angles are CONGRUENT (equal). They are not supplementary.

B. SHOW CORRECT WORK (1 POINT)

$3x + 10 = 5x - 20$

$30 = 2x$

$x = 15$

Each angle = $3(15) + 10 = 55°$

2. Linear pair

ORIGINAL PROBLEM

Adjacent angles on a straight line measure $(2x + 20)°$ and $(x + 40)°$. Find x.

INCORRECT WORKED SOLUTION

$2x + 20 = x + 40$

$x = 20$

A. EXPLAIN THE MISTAKE (1 POINT)

Angles on a straight line add to 180°. They are not necessarily equal.

B. SHOW CORRECT WORK (1 POINT)

$(2x + 20) + (x + 40) = 180$

$3x + 60 = 180$

$x = 40$

Angles: 100° and 80°

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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3. Corresponding angles

ORIGINAL PROBLEM

a ∥ b. Corresponding angles measure (3x + 5)° and (5x − 25)°. Find x.

INCORRECT WORKED SOLUTION

(3x + 5) + (5x − 25) = 180

8x − 20 = 180

8x = 200

x = 25

A. EXPLAIN THE MISTAKE (1 POINT)

Corresponding angles formed by PARALLEL lines are CONGRUENT, not supplementary.

B. SHOW CORRECT WORK (1 POINT)

3x + 5 = 5x − 25

30 = 2x

x = 15

Each angle = 3(15) + 5 = 50°

4. Same-side interior angles

ORIGINAL PROBLEM

a ∥ b. Same-side interior angles measure (3x + 15)° and (2x + 10)°. Find x.

INCORRECT WORKED SOLUTION

3x + 15 = 2x + 10

x = −5

A. EXPLAIN THE MISTAKE (1 POINT)

Same-side interior angles are SUPPLEMENTARY (they add to 180°). They are not equal. (A negative x here would give negative angles.)

B. SHOW CORRECT WORK (1 POINT)

(3x + 15) + (2x + 10) = 180

5x + 25 = 180

5x = 155

x = 31

Angles: 108° and 72°

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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5. Proving lines parallel

ORIGINAL PROBLEM

Given: ∠1 ≅ ∠2 (alternate interior angles).
Prove: a ∥ b.

INCORRECT WORKED SOLUTION

1. a ∥ b
2. Alternate interior angles are congruent.
3. ∠1 ≅ ∠2

A. EXPLAIN THE MISTAKE (1 POINT)

This proof ASSUMES what must be proven (a ∥ b) and ends with the Given. Start from ∠1 ≅ ∠2 and use the CONVERSE of the Alternate Interior Angles Theorem.

B. SHOW CORRECT WORK (1 POINT)

1. ∠1 ≅ ∠2 (Given)
2. If alternate interior angles are congruent, the lines are parallel (Converse of the Alternate Interior Angles Theorem)
3. a ∥ b

6. Angle bisector

ORIGINAL PROBLEM

BD bisects ∠ABC. m∠ABC = 100° and m∠ABD = (4x + 10)°. Find x.

INCORRECT WORKED SOLUTION

4x + 10 = 100

4x = 90

x = 22.5

A. EXPLAIN THE MISTAKE (1 POINT)

A bisector splits the angle into TWO EQUAL parts. ∠ABD is HALF of 100°, so it is 50°.

B. SHOW CORRECT WORK (1 POINT)

$m\angle ABD = \frac{100}{2} = 50°$

4x + 10 = 50

4x = 40

x = 10

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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7. Complement word problem

ORIGINAL PROBLEM

An angle measures 20° more than its complement. Find the angle.

INCORRECT WORKED SOLUTION

Let x = the complement. The angle is x + 20.

x + (x + 20) = 180

2x = 160

x = 80, so the angle is 100°.

A. EXPLAIN THE MISTAKE (1 POINT)

Complementary angles add to 90°, not 180°.

B. SHOW CORRECT WORK (1 POINT)

x + (x + 20) = 90

2x = 70

x = 35 (the complement)

The angle is 35 + 20 = 55°.

8. Solving carefully

ORIGINAL PROBLEM

a ∥ b. Alternate interior angles measure (7x − 9)° and (4x + 21)°. Find x and the angle measure.

INCORRECT WORKED SOLUTION

7x − 9 = 4x + 21

3x − 9 = 21

3x = 12

x = 4

A. EXPLAIN THE MISTAKE (1 POINT)

Adding 9 to both sides of 3x − 9 = 21 gives 3x = 30, not 12. (The wrong answer even makes the angles unequal.)

B. SHOW CORRECT WORK (1 POINT)

7x − 9 = 4x + 21

3x − 9 = 21

3x = 30

x = 10

Each angle: 7(10) − 9 = 61°

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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9. Angles around a point

ORIGINAL PROBLEM

Three angles around a point measure (2x)°, (3x + 10)° and (x + 50)°. Find x.

INCORRECT WORKED SOLUTION

2x + 3x + 10 + x + 50 = 180

6x + 60 = 180

6x = 120

x = 20

A. EXPLAIN THE MISTAKE (1 POINT)

Angles around a point add to 360° (a full turn). A sum of 180° is for a straight line.

B. SHOW CORRECT WORK (1 POINT)

2x + (3x + 10) + (x + 50) = 360

6x + 60 = 360

6x = 300

x = 50

Angles: 100°, 160°, 100°

10. Naming angle pairs

ORIGINAL PROBLEM

a ∥ b with transversal t. ∠4 and ∠5 are between the lines on the same side of t. m∠4 = 112°. Find m∠5.

INCORRECT WORKED SOLUTION

∠4 and ∠5 are alternate interior angles.

Alternate interior angles are congruent.

m∠5 = 112°

A. EXPLAIN THE MISTAKE (1 POINT)

Angles between the lines on the SAME side of the transversal are same-side interior angles, which are SUPPLEMENTARY: m∠5 = 180° − 112° = 68°.

B. SHOW CORRECT WORK (1 POINT)

∠4 and ∠5 are same-side interior angles.

Same-side interior angles are supplementary.

m∠5 = 180° − 112° = 68°

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points